Australia’s national diagnostic network for aquatic animal diseases has been developed over time to support the many functions of Australia’s aquatic animal health management system. These include:

- confirmation or exclusion of exotic diseases
- implementing disease management measures
- enterprise health accreditation
- demonstration of regional or national disease status.

Australia’s diagnostic network

Australia’s diagnostic network draws on nodes of expertise throughout the national, state and Northern Territory government laboratories, research laboratories and private service providers. This high standard of diagnostic service is built upon:

- quality research
- validation of methods
- diagnostic and laboratory standards
- programs to support quality assurance.

National reference laboratory for aquatic animal diseases

The CSIRO Australian Centre for Disease Preparedness is the national reference laboratory for aquatic animal diseases and provides services for exclusion testing for significant diseases.

Further information on Australia’s animal health laboratories, including laboratory policies and plans, laboratory tests, point-of-care tests, and reference laboratories and collaborating centres, is available on the department’s website.

Field and laboratory diagnostic resources

A range of field and diagnostic resources have been developed to ensure Australia maintains the capacity to identify and diagnose aquatic animal diseases of national significance.

The Aquatic Animal Diseases Significant to Australia: Identification Field Guide aims to help people recognise diseases of significance to aquaculture and fisheries in Australia. The 5th edition of the field guide is the most recent revision and incorporates information on 53 aquatic animal diseases of finfish, crustaceans, molluscs and amphibians.

The Department of Agriculture, Fisheries and Forestry (DAFF) maintains the list of Australian and New Zealand standard diagnostic procedures (ANZSDPs). ANZSDPs describe standardised diagnostic procedures that have optimal accuracy, robustness, sensitivity and specificity for animal health diagnostic laboratories in Australia and New Zealand. They seek to provide consistency between laboratories and develop a quality assurance system that includes proficiency testing programs. ANZSDPs are available for both terrestrial and aquatic animal diseases.

Aquatic ANZSDPs are developed to be consistent with the World Organisation for Animal Health (WOAH) Manual of Diagnostic Tests for Aquatic Animals (Aquatic Manual) but may exceed those requirements where special procedures and/or interpretation is necessary for Australian and New Zealand circumstances.

Where an ANZSDP has not been developed for a particular test, the methods recommended in the WOAH Aquatic Manual should be used.

National diagnostic capability and capacity building projects

The department, in collaboration with state and territory governments and industry, coordinates activities to manage aquatic animal health nationally. This includes activities to enhance Australia's aquatic animal health diagnostic system. These projects are outlined below.

Status: Ongoing.

Project: 2022–2025

The Australian proficiency testing (PT) program for aquatic animal disease enables Australian laboratories to assess their diagnostic capabilities to correctly detect priority aquatic animal diseases using molecular methods. Participants include private, university and state/territory government laboratories. Participating laboratories benefit from benchmarking to support reproducibility and validation of tests, strengthening competencies and laboratory techniques, and support for accreditation.

Australia's government and private laboratories can participate in PT for 8 aquatic animal disease agents:

- haliotid herpesvirus-1 (HaHV-1)
- yellowhead virus-1 (YHV1)
- ostreid herpesvirus-1 microvariant (OsHV-1)
- nervous necrosis virus (NNV)
- white spot syndrome virus (WSSV)
- megalocytiviruses (MCV)
- Bonamia exitiosa
- Perkinsus olseni.

The national PT program is run by the CSIRO Australian Centre for Disease Preparedness ISO accredited PT scheme provider. Information on the regional PT program is available on the International activities webpage.

Status: Ongoing.

Neptune is Australia's aquatic animal disease information management system, collaboratively developed by the CSIRO and the department. It is an online database of all published reports of aquatic animal diseases and pathogens in Australia, some of which contain high resolution whole-slide digital images of histopathology. Neptune will be made accessible to interested stakeholders as a tool for sharing aquatic animal health knowledge.

Australia's aquatic animal community, including national, state and territory governments, aquatic animal industry, researchers and other aquatic animal health professionals, have all contributed to the information found on Neptune.

Next steps in Neptune's development have been included in AQUAPLAN 2022-2027.

Status: Ongoing.

Pooling of samples (for example using 5 samples and testing them as one) is a useful technique for reducing testing costs and increasing sampling intensity. However, by its nature, pooling could potentially dilute very low numbers of pathogens in samples to below detectable levels, reducing sensitivity. In this project, the effect of pooling on diagnostic sensitivity of tests was evaluated for the purposes of surveillance (including sub-clinical infection) of white spot syndrome virus (WSSV), yellow head virus-1 (YHV-1) and megalocytiviruses, all of which are aquatic animal pathogens of national and trade significance. The intended outcome was to determine the effect of pooling on diagnostic sensitivity, enable testing to be more cost-effective, and increase confidence in result accuracy.

A summary of the project results will be provided here once completed.

Status: WSSV paper published (see below). OsHV-1 paper yet to be published.

Validation of diagnostic tests for aquatic animal diseases has been recognised by the World Organisation for Animal Health (WOAH) as necessary to improve confidence in test results and to provide evidence required for accurate interpretation of laboratory test results.

Two studies were conducted through this project, one for white spot syndrome virus (WSSV) and one for ostreid herpesvirus-1 (OsHV-1).

The WSSV study aimed to generate data on performance characteristics for 2 real-time TaqMan PCR assays (CSIRO and WOAH WSSV qPCRs) for the purposes of (1) detection of WSSV in clinically diseased prawns and (2) detection of WSSV in apparently healthy prawns. The assays demonstrated comparable performance characteristics, and the results contributed to the validation data required in the WOAH validation pathway for the purposes of detection of WSSV in clinically diseased and apparently healthy prawns.

The OsHV-1 study is yet to be published. Information on that study and a link to the paper will be provided once complete.

Published papers:

Article	Access
Moody NJG, Mohr PG, Williams LM, Cummins DM, and others. (2022) Performance characteristics of two real-time TaqMan polymerase chain reaction assays for the detection of WSSV in clinically diseased and apparently healthy prawns. Diseases of Aquatic Organisms, 150:169-182.	https://doi.org/10.3354/dao03687

The published paper for OsHV-1 will be provided here once complete.

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